

**Institute of Information Technology
University of Dhaka**

**Topic: Goal Question Metrics(GQM)
Software Metrics (SE-611)**

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1. Introduction

GQM is the abbreviation for "goal, question, metric". It is an established goal-oriented approach to software metrics to improve and measure software quality.

GQM defines a measurement on three dimensions. They are–

Conceptual Level (Goal): A goal is defined for an object, for a variety of reasons, with respect to various models of quality, from various points of view and relative to a particular environment.

Operational level (Question): A set of questions is used to define models of the object of study and then focuses on that object to characterize the assessment or achievement of a specific goal.

Quantitative level (Metric): A set of metrics, based on the models, is associated with every question in order to answer it in a measurable way.

The first step is deciding our goal, then develop a question set to collect data from stakeholders. The questions must have appropriate metrics to measure the result later. We have to create a questionnaire to collect data from people defined in the scope. The last part will include metrics analysis with result derivation to finally reach an outcome.

2. Project Specification

2.1 Project Overview

Our objective is to know students' perceptions and ethical concerns regarding AI tools by measuring the frequency and variety of AI tools, the perceived impact of AI tools on their learning outcomes, academic performance, and personal integrity.

2.2 Motivation

The AI usage boom in our daily life started with chatgpt. Subsequently we have come across diverse AI tools and use them in our day to day life. With this new technology, age old ethical issues are re-emerging with new forms. What are the users thinking about AI's ethical issues? Are they actually learning through the AI tools? How are AI tools impacting students' academic performance? Are they confident about their academic integrity? These were the fundamental questions driving us for this project.

2.3 Scope

We have reached out to both undergraduate and postgraduate students, although most of the respondents are undergraduate level students.

3. Goal Specification

3.1 GQM Framework

General Statement: Investigating students' Perceptions and Ethical Concerns Regarding AI Tools by Measuring the Frequency and Variety of AI Tools, the Perceived Impact of AI Tools on Their Learning Outcomes, Academic Performance, and Personal Integrity: An Analysis.

3.2 PPE Approach

Purpose: To evaluate whether students are concerned about ethical aspects of LLMs and other AI tools with factual information to determine the perceptual impact on learning outcomes, academic performance and personal integrity.

Perspective: To examine the variety and frequency of AI tool usage to understand the perception regarding ethical impacts.

Environment: We want to understand how students from a diverse set of curriculums rely on AI tools in an ethically acceptable way.

After specifying our goal through the purpose-perspective-environment framework, our final goal is –

“To examine the variety and frequency of AI tools in order to understand the perception regarding learning outcomes, academic performance and personal integrity from the viewpoint of students.”

3.3 Sub Goals

Our goal has 5 more sub goals:

- A. Evaluate the range of AI tool usage.
- B. Determine the motivating factors for relying on AI tools.
- C. Figuring out the obstacles faced from AI tools.
- D. Finding out trust in LLMs compared to teachers.
- E. Knowing the level of accessibility of AI tools for Bangladeshi students.

3.4 Question and metrics

Q1: Are students aware of other AI tools besides ChatGPT?

- **M1:** Percentage of surveyed students who can identify at least three AI tools other than ChatGPT.
- **M2:** Percentage of surveyed students who can correctly describe the primary use case for at least one AI tool other than ChatGPT.

- **M3:** Frequency of mentions of other AI tools in the "Other" field provided by students.

Q2: What tasks do students use AI tools for?

- **M1:** Percentage of surveyed students selecting each task category (e.g., "For research and literature studies," "For programming and simulations").
- **M2:** Number of unique "Other" tasks entered by students that indicate creative or novel use cases.
- **M3:** Correlation between task categories selected and the diversity of AI tools mentioned by students.

Q3: How reliant are students on AI tools compared to traditional resources (e.g., books, web)?

- **M1:** Average rating (on a scale of 1 to 5) for the statement "When searching for something, I tend to use AI tools more than web or book resources."
- **M2:** Percentage of students rating their reliance on AI for idea brainstorming at 4 or 5.
- **M3:** Distribution of responses for reliance on AI tools for decision-making tasks.

Subgoal B. Determine the motivating factors for relying on AI tools.

Q1: What motivates students to use AI tools in their work?

- **M1:** Percentage of surveyed students selecting each motivating factor (e.g., curiosity, academic pressure, time constraints).
- **M2:** Number of unique "Other" motivations mentioned by students.
- **M3:** Correlation between selected motivating factors and frequency of AI tool usage.

Q2: What features of AI tools do students find most valuable?

- **M1:** Percentage of students selecting each feature (e.g., velocity, comprehensibility, usability).
- **M2:** Frequency of mentions of unique "Other" features provided by students.
- **M3:** Average feature rating (calculated based on the total number of features selected by each student).

Q3: How impactful are AI tools in reducing work time and improving productivity?

- **M1:** Average rating (on a scale of 1 to 5) for the statement "The use of AI tools reduced the time for me to understand a topic."
- **M2:** Percentage of students rating their agreement with "AI tools enable me to do my work significantly faster" at 4 or 5.
- **M3:** Percentage of students rating their agreement with "There were tasks that I would not have been able to complete without the use of AI tools" at 4 or 5.

Subgoal C. Figuring out the obstacles faced from AI tools.

Q1: What are the most common obstacles students face when using AI tools?

- **M1:** Percentage of surveyed students selecting each problem (e.g., incorrect answers, biased answers, bad reasoning).
- **M2:** Frequency of mentions of unique "Other" obstacles provided by students.
- **M3:** Correlation between specific obstacles (e.g., hallucination or paywalls) and lower usage frequency of AI tools.

Q2: How frequently do students encounter incorrect or inconsistent responses from AI tools?

- **M1:** Average rating (on a scale of 1 to 5) for the statement "I have often got wrong/inconsistent answers from AI tools."
- **M2:** Percentage of students rating their agreement with this statement at 4 or 5.
- **M3:** Frequency of follow-up actions (e.g., re-prompting or abandoning the tool) due to incorrect answers, as reported in open-ended responses.

Q3: Are students concerned about privacy and data tracking by AI tools?

- **M1:** Average rating (on a scale of 1 to 5) for the statement "I am concerned that many of the AI tools are storing and tracking my personal information."
- **M2:** Percentage of students rating their concern at 4 or 5.
- **M3:** Percentage of students who cite privacy concerns as a reason for limiting their AI tool usage.

Subgoal D. Finding out trust in LLMs compared to teachers.

Q1: How frequently do students rely on AI tools for understanding complex topics in their coursework?

- **M1:** Percentage of students selecting each frequency option (Always, Often, Sometimes, Rarely, Never).
- **M2:** Correlation between frequency of reliance on AI tools and students' reported trust in AI tools compared to teachers.

Q2: Which source do students trust more for understanding difficult concepts?

- **M1:** Percentage of students choosing "Primarily AI tools," "Primarily my teachers," "Equally AI tools and teachers," or "Neither."
- **M2:** Comparison of trust levels across different academic disciplines or coursework types.

Q3: How confident are students in the accuracy of answers from AI tools compared to their teachers?

- **M1:** Distribution of responses across confidence levels ("Much more confident in AI tools" to "Much more confident in teachers").
- **M2:** Percentage of students rating confidence in AI tools equal to or greater than teachers.

Q4: What do students do when AI tools and teachers provide conflicting explanations?

- **M1:** Percentage of students choosing each response option ("The AI tool," "The teacher," "Further research," "I'm not sure").
- **M2:** Frequency of students choosing "Further research" as an approach to resolve conflicts.

Q5: Do students perceive AI tools as offering more unbiased information compared to teachers?

- **M1:** Average rating (on a scale of 1 to 5) for the statement "AI tools offer more unbiased information than teachers."
- **M2:** Percentage of students agreeing (rating 4 or 5) with this statement.

Q6: How well do students think AI tools understand the context and requirements of their assignments compared to teachers?

- **M1:** Distribution of responses across perceived understanding levels ("Much better than teachers" to "Much worse than teachers").

- **M2:** Percentage of students rating AI tools as equal to or better than teachers in understanding context.

E: Accessibility of AI Tools for Bangladeshi Students

Q1: What type of internet access do students primarily use for AI tools?

- **M1:** Percentage of students using mobile data, broadband/home Wi-Fi, public Wi-Fi, or other sources.
- **M2:** Correlation between type of internet access and frequency of AI tool usage.

Q2: How often do technical issues affect students' use of AI tools?

- **M1:** Average rating (on a scale of 1 to 5) for the frequency of technical issues.
- **M2:** Percentage of students reporting "Very Often" or "Never" facing technical issues.
- **M3:** Commonly reported technical challenges (e.g., slow network, electricity outages).

Q3: Does the cost of internet affect students' AI tool usage?

- **M1:** Percentage of students who report "Always" or "Never" being influenced by internet costs.
- **M2:** Correlation between internet costs and frequency of AI tool usage.
- **M3:** Percentage of students who have avoided AI tools due to high internet costs.

Q4: What type of AI subscription packages do students use?

- **M1:** Percentage of students using free, paid, shared, or other subscription types.
- **M2:** Distribution of students sharing accounts versus individual subscriptions.

Q5: How do students perceive the affordability of premium AI services?

- **M1:** Average affordability rating (on a scale of 1 to 5).
- **M2:** Percentage of students rating premium services as "Very Affordable" or "Very Costly."

- **M3:** Likelihood of students opting for free/open-source tools due to affordability issues.

Q6: How do internet outages affect students' AI-based tasks?

- **M1:** Percentage of students reporting "No effect," "Slight delay," "Significant delay," or "Task abandonment."
- **M2:** Correlation between internet outages and reduced assignment completion rates.

4. Questionnaire Preparation and Data Collection

This report was prepared with data collected from undergraduate and postgraduate students from various universities across Bangladesh. The report does not contain data from people who never attended university.

To facilitate our investigation, we carefully developed this survey questionnaire so that it covered a wide spectrum of topics, all tied to the usage of AI tools.

Demographic Information: The size of demographic data collected from respondents is small compared to various aspects of AI tools. The demographic information includes their academic majors and level of education (undergraduate or postgraduate). The report mainly focuses on the ethical aspects of AI tools, as well as its holistic impacts.

Some sample questions are given below–

1. **I use AI tools for the following tasks: (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter) ***
 - For regular study
 - For research and literature studies
 - For concept development
 - For creating design, images and art

- For data analysis, data visualisation, modelling
- For problem solving, decision making
- For text analysis, text processing, text creation
- For translations
- For understanding a novel topic
- For entertainment
- For language processing (e.g . Paraphrasing, proofreading)
- For exam preparation
- For programming and simulations
- Other:

2. **What do you think are the best features of AI tools? (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter) ***

- Velocity (Quickly generates the content I want)
- Comprehensibility (Helps to explain topics easily)
- Usability (Very easy and intuitive to use)
- Availability (The tools are free to use)
- Extensibility (Helps me to complete more tasks within time)
- Capability (The tools can generate almost anything that I ask for)
- Other:

Full questionnaire can be found here: <https://forms.gle/9kB6iw7opycdBfL8A>

5. Data Visualization

The summary of the collected data is presented here in graphs and charts:

Demographic:

Academic Major Discipline

58 responses

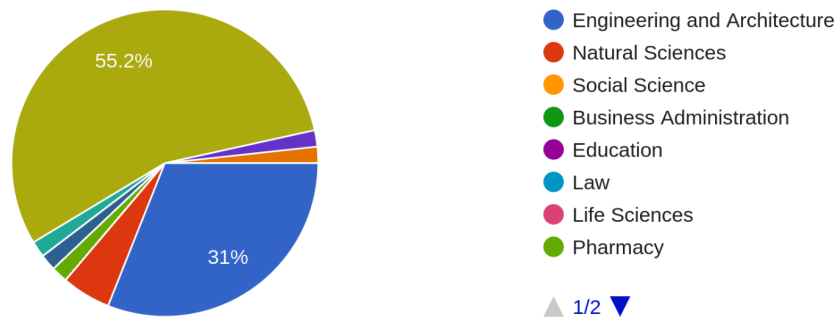


Fig 1.1: Respondents' academic major

Academic Major Discipline

58 responses

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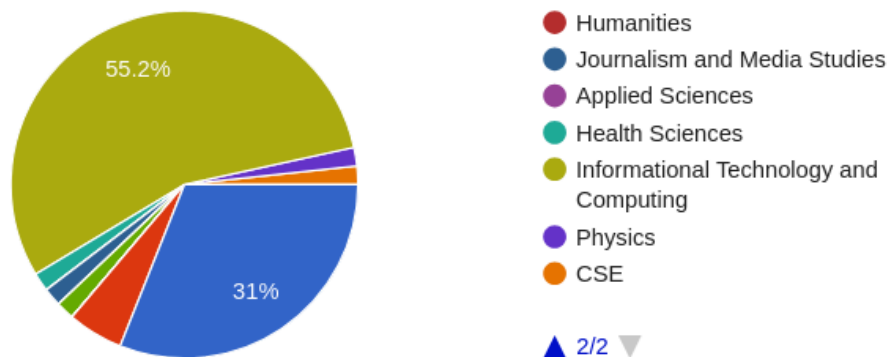


Fig 1.2: Respondents' academic major

What is your Educational Level

58 responses

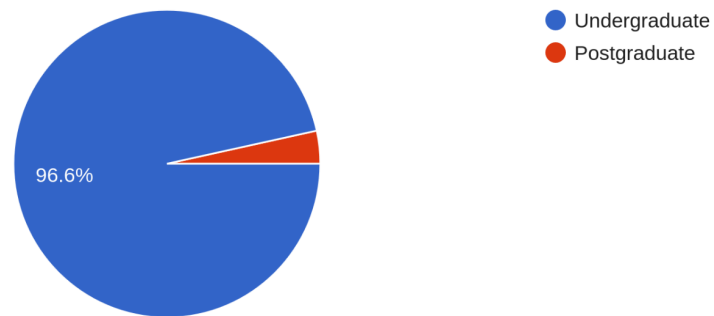


Fig 2: Respondents' level of education

The Use of AI Tools:

Are you familiar with AI tools?

58 responses

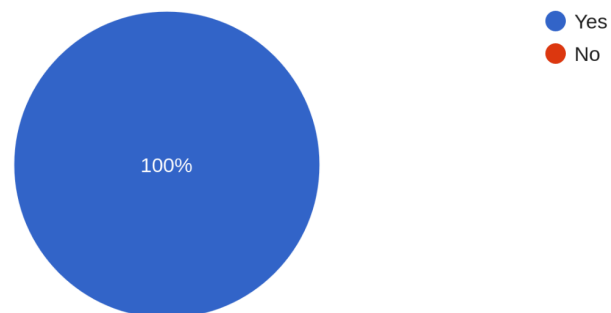


Fig 3: Respondents' familiarity with AI

Which AI tools are you familiar with? (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter)

58 responses

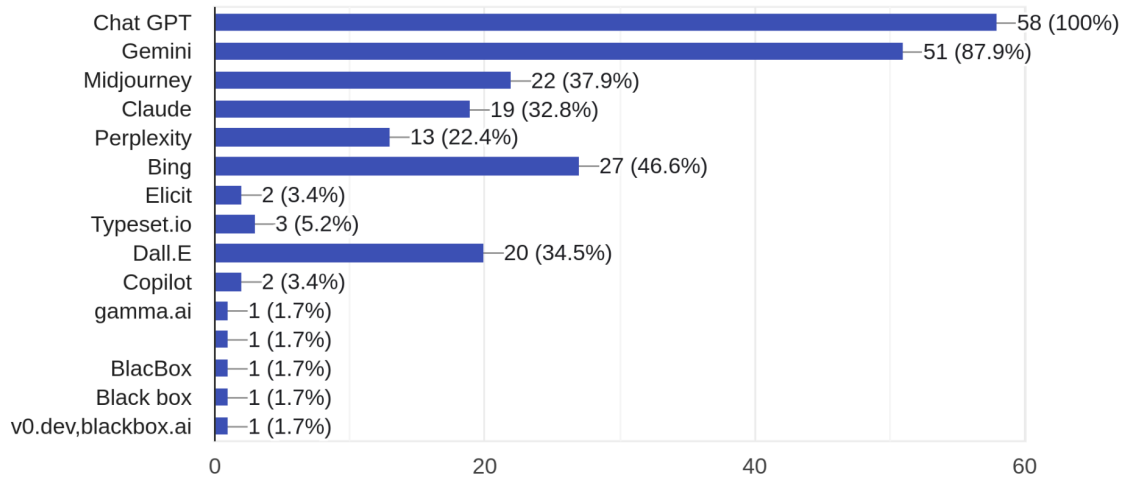


Fig 4: Respondents' familiarity with AI tools

I use AI tools for the following tasks: (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter)

58 responses

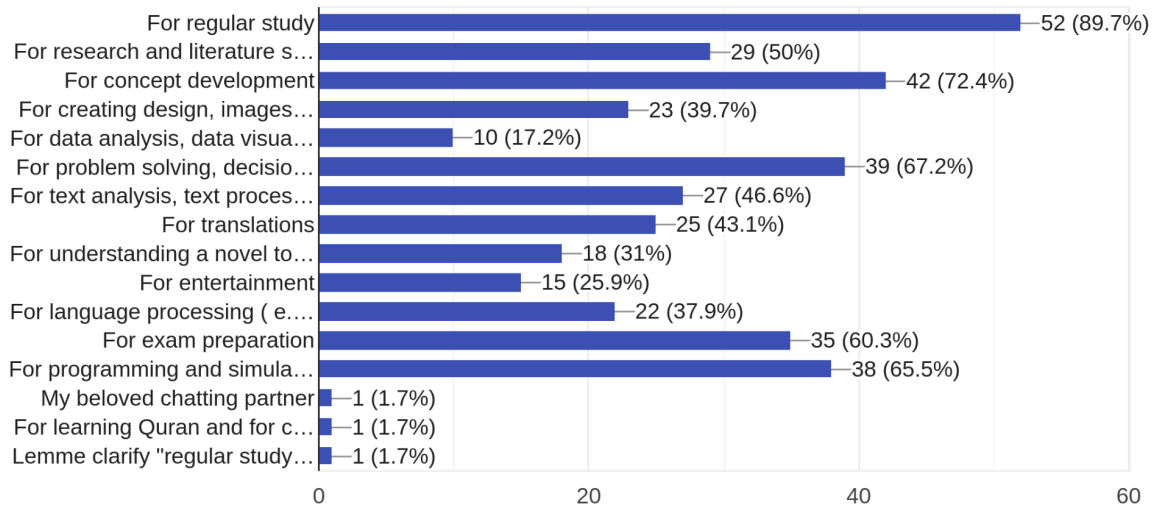


Fig 5: Respondents' reasons for using AI tools

I regularly use/used AI for my studies and academic works

58 responses

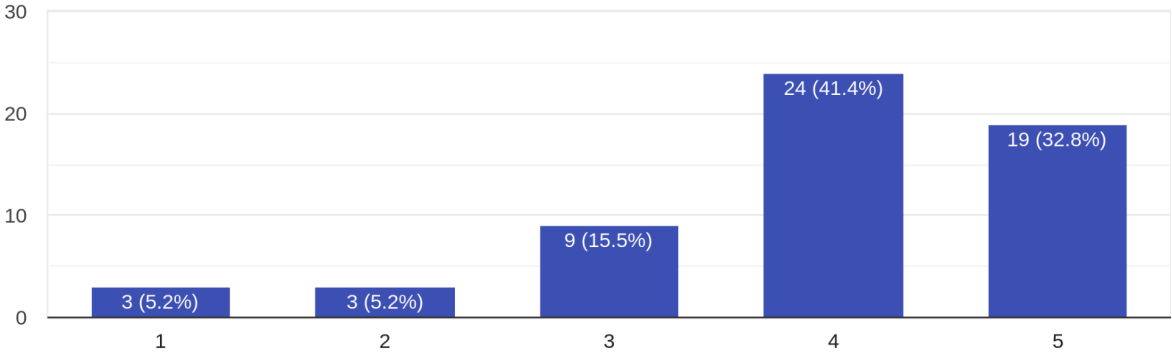


Fig 6: Respondents’ AI usage for studies and academic work

When searching for something, I tend to use AI tools more than web or book resources

58 responses

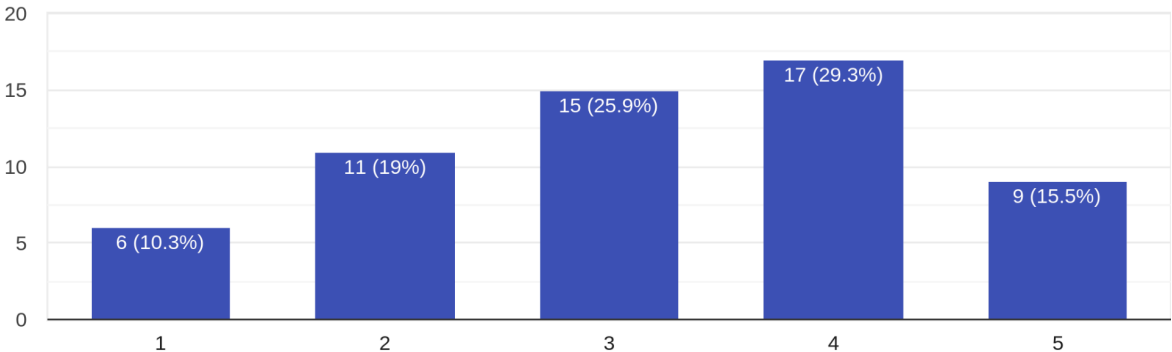


Fig 7: Respondents’ usage of AI tools for searching

When brainstorming for any ideas, I tend to use AI for that purpose

58 responses

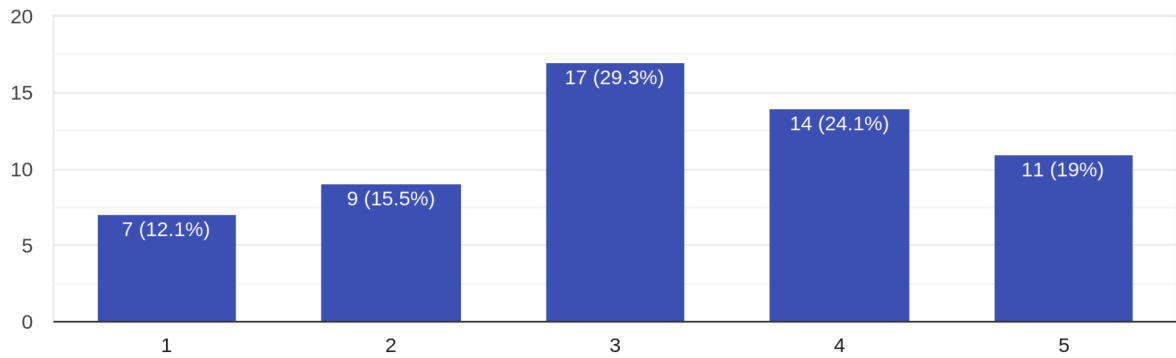


Fig 8: Respondents' usage of AI tools for brainstorming

I often use AI for decision-making tasks

58 responses

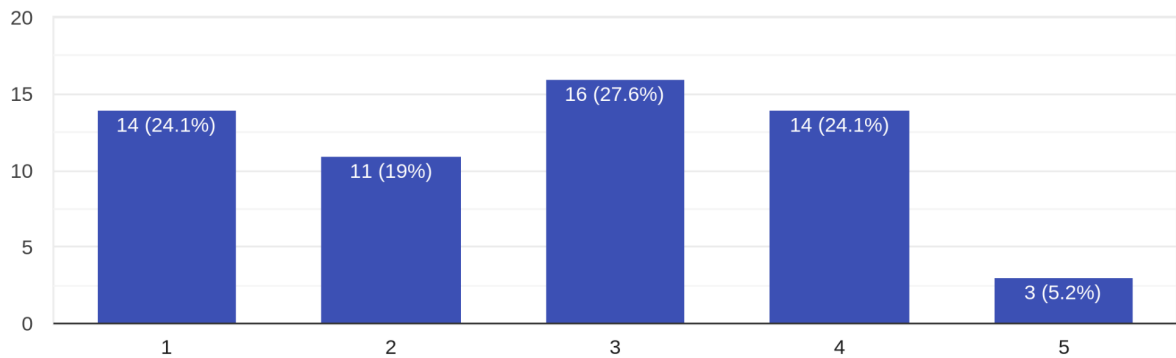


Fig 9: Respondents' usage of AI tools for decision making

I regularly use AI plugins in the software tools I use (e.g. Copilot in VS Code, Content generating AI in Canva)

58 responses

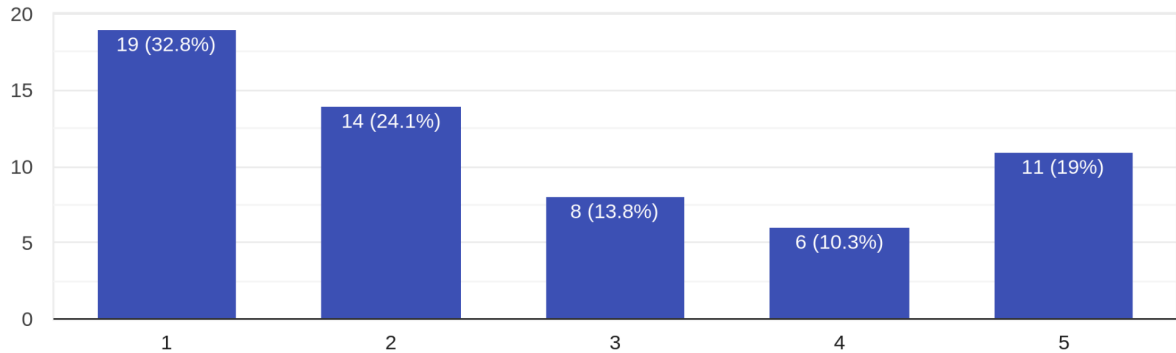


Fig 10: Respondents' usage of AI plugins

Motivating Factor for AI:

What motivated you to use AI tools in your work? (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter)

58 responses

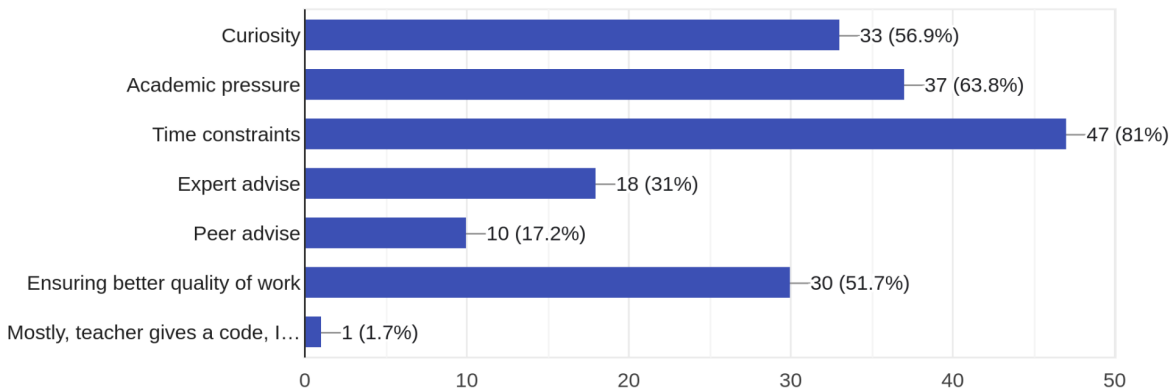


Fig 11: Respondents' motivation for using AI tools

What do you think are the best feature of AI tools? (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter)

58 responses

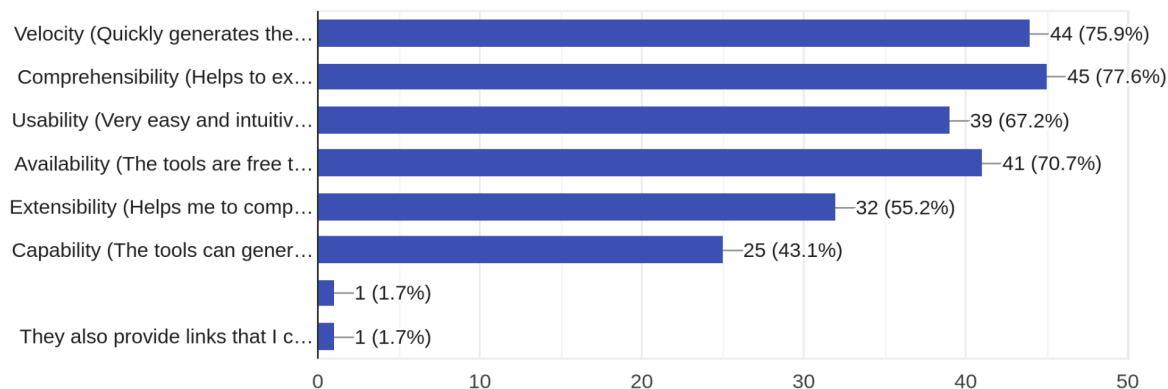


Fig 12: Respondents' response to best features

The use of AI tools reduced the time for me to understand a topic

58 responses

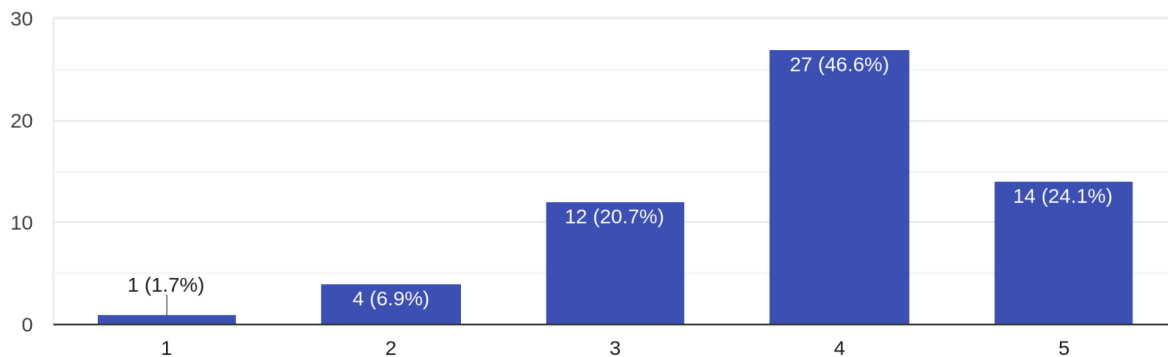


Fig 13: Respondents' response to reduced time

AI tools enables me to do my work significantly faster

58 responses

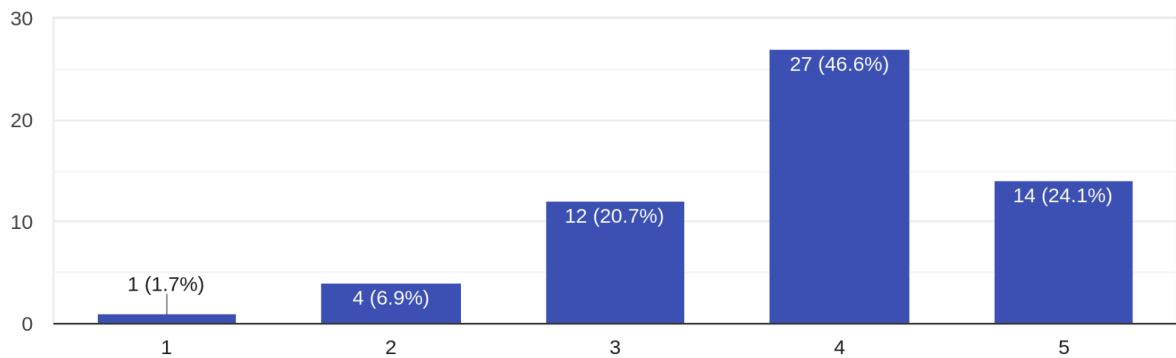


Fig 14: Respondents' response for working faster

There were tasks that I would not have been able to complete without the use of AI tools

58 responses

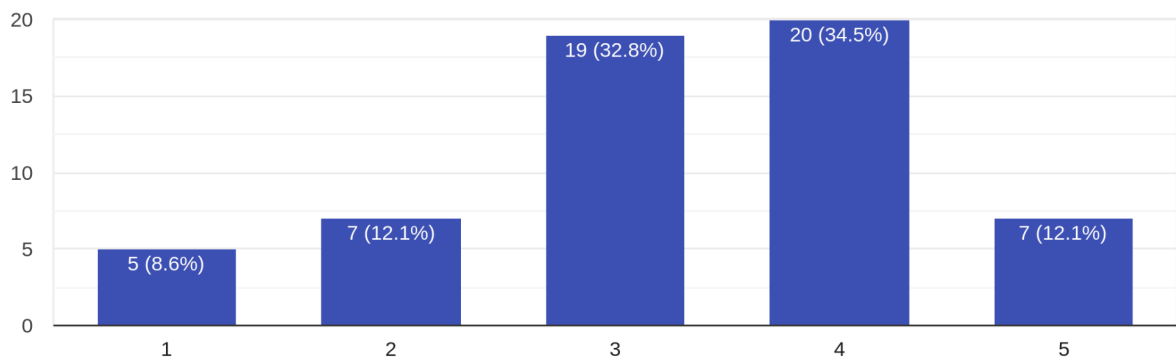


Fig 15: Respondents' response for completing tasks

The Obstacles of AI Tools:

Which problems have you faced when using AI tools? (Select all that applies. You may also enter your own preferences in "other". Write them with comma (,) if you have multiple preferences to enter)

58 responses

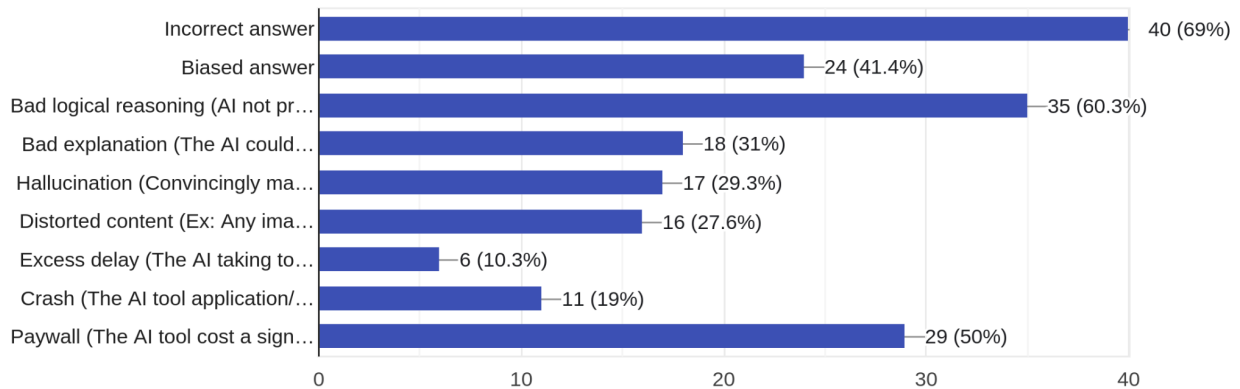


Fig 16: Respondents' problems with AI tools

I have often got wrong/inconsistent answers from AI tools

58 responses

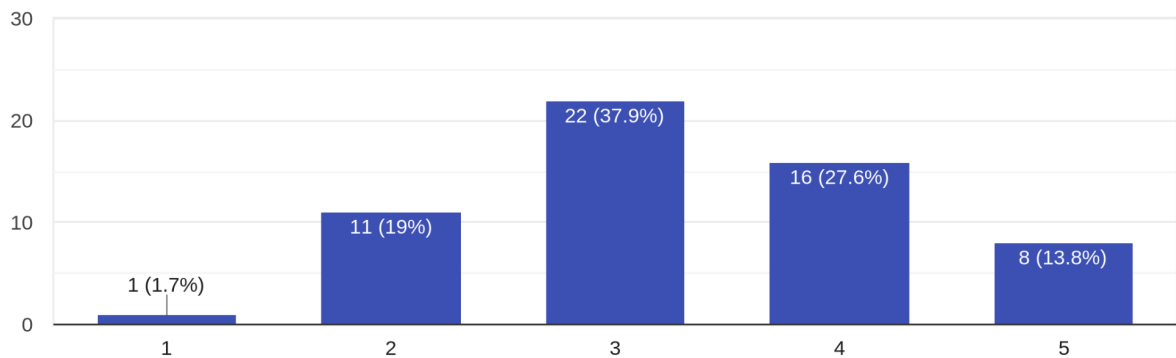


Fig 17: Respondents' response for getting wrong answers

I am concerned that many of the AI tools are storing and tracking my personal information

58 responses

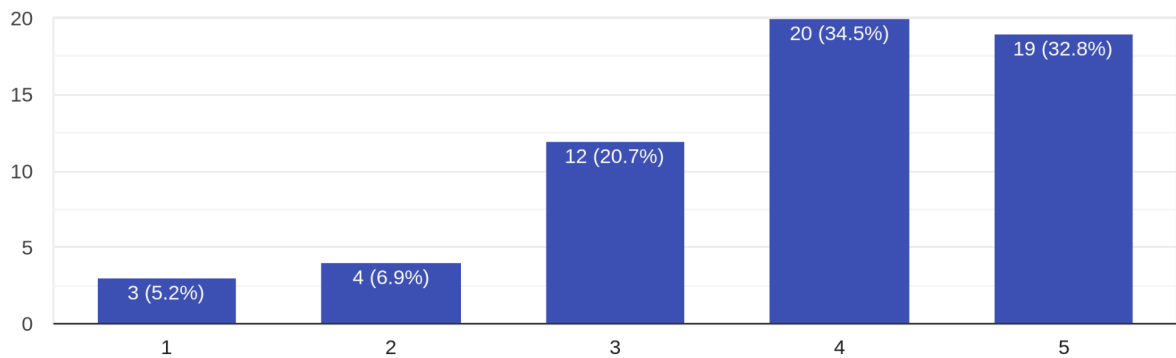


Fig 18: Respondents' concern for personal information

The Ethical Aspects

I am aware about my institute's academic honesty policy

58 responses

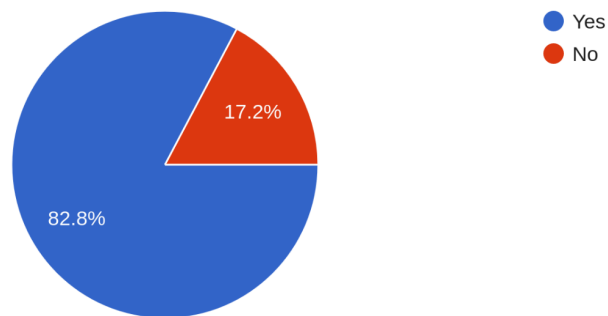


Fig 19: Respondents' awareness of honesty policy

My institution's academic honesty policy is against AI tools

58 responses

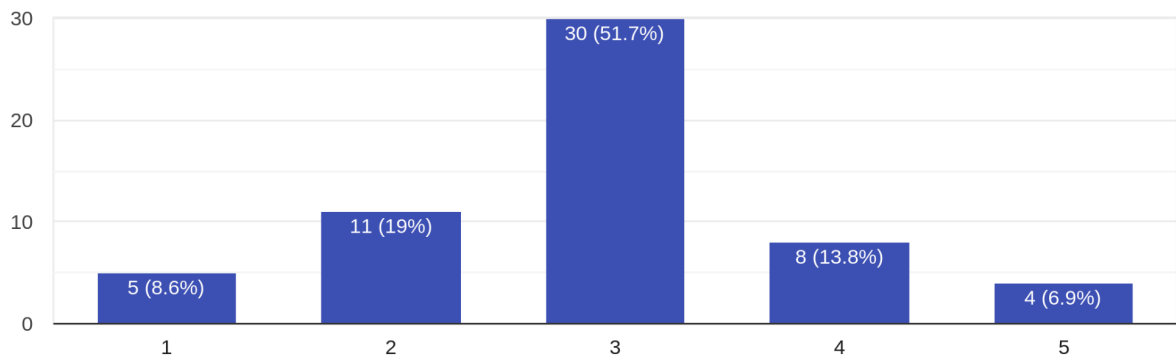


Fig 20: Respondents' view of academic honesty policy

My teachers/instructors discussed about the ethical usage of various AI tools in their classes

58 responses

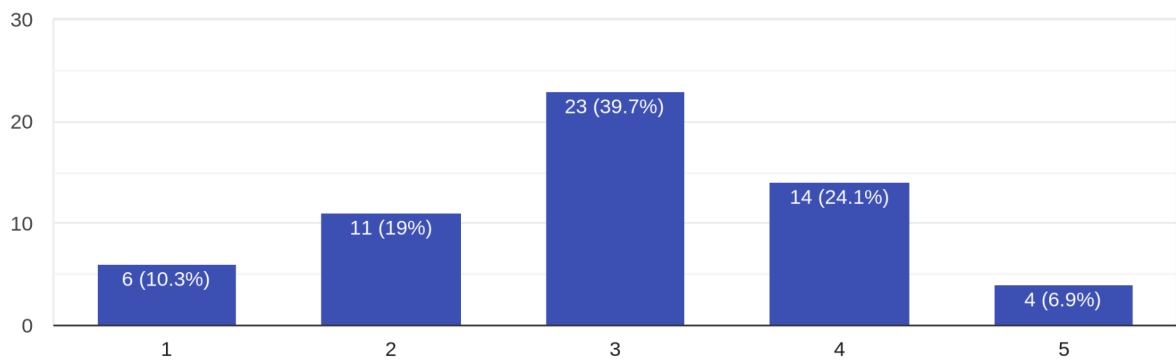


Fig 21: Respondents' responses regarding discussion in classes

My teachers/instructors are aware of my use of AI tools in their tasks

58 responses

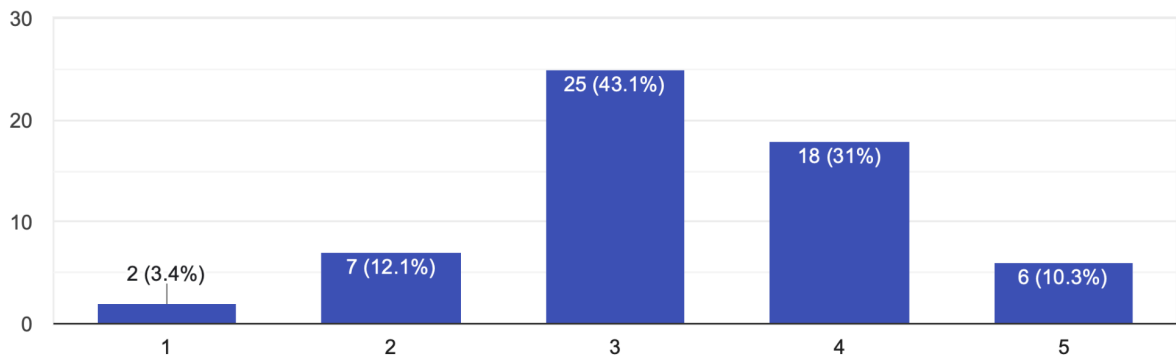


Fig 22: Respondents' response to teachers' awareness

How likely are you to continue using AI tools even if your teachers advise against them?

58 responses

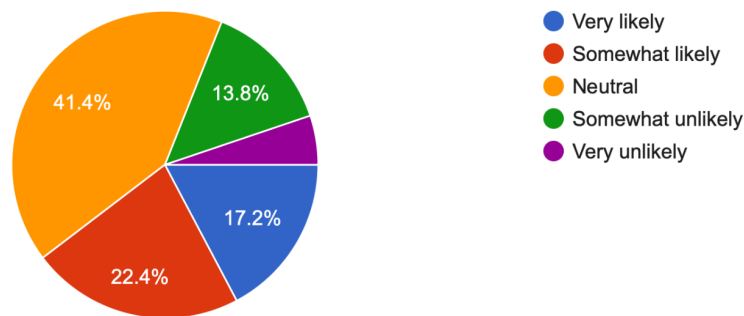


Fig 23: Respondents' likelihood of continuation

I think that students should be allowed to use AI tools for their assignments and lab tasks

58 responses

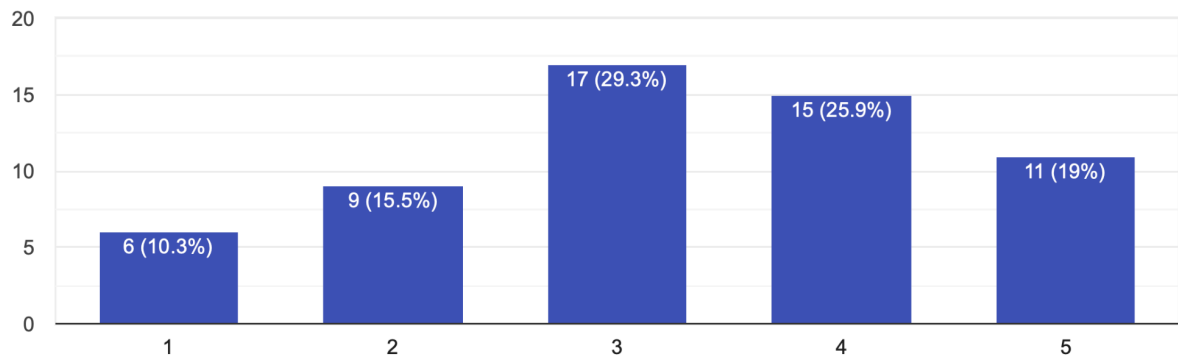


Fig 24: Respondents' view of allowing AI tools

I think that paraphrasing, proof-reading and summarization work for written assignments are fair use cases for AI

58 responses

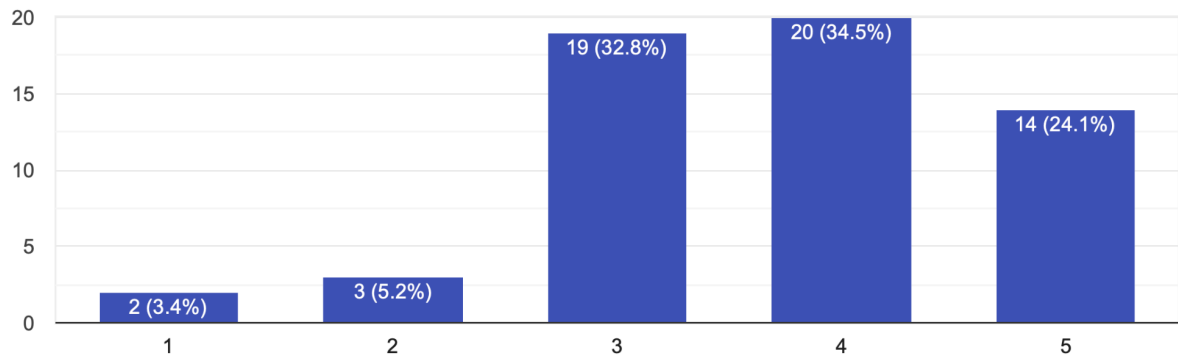


Fig 25: Respondents' view of fair use cases

I have often used AI tools in some academic activities where it was not allowed

58 responses

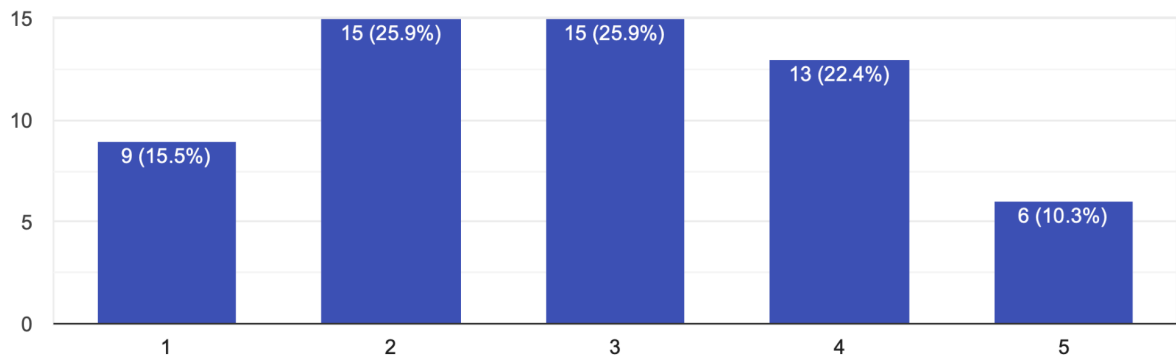


Fig 26: Respondents' usage of AI in disallowed cases

Which factors drove you to use AI unethically in your academic activities? (Select all that applies.

You may also enter your own preferences in "other".....mma (,) if you have multiple preferences to enter)

58 responses

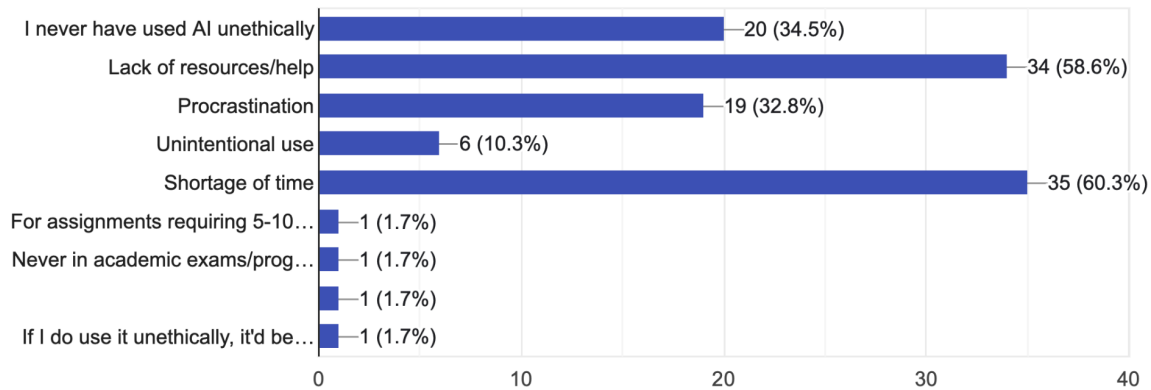


Fig 27: Respondents' factors for unethically using AI

I was often penalized for the use of AI tools in academic activities

58 responses

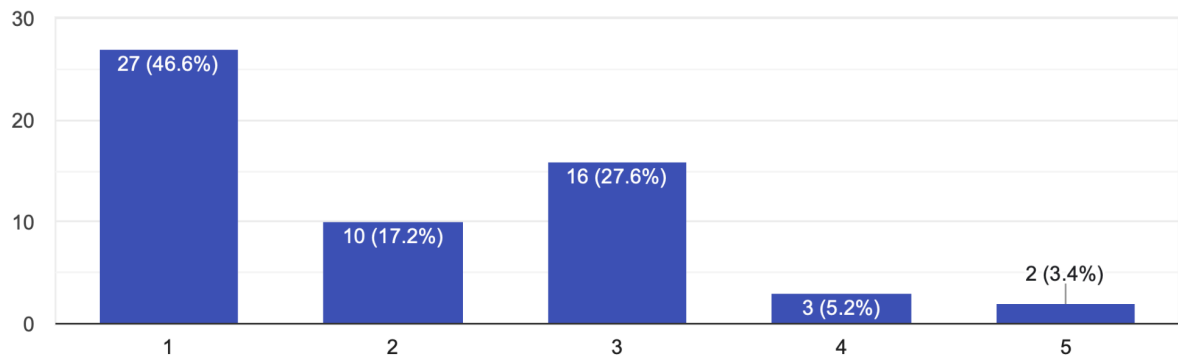


Fig 28: Respondents' cases of being penalized

I view AI generated content as copies of original works

58 responses

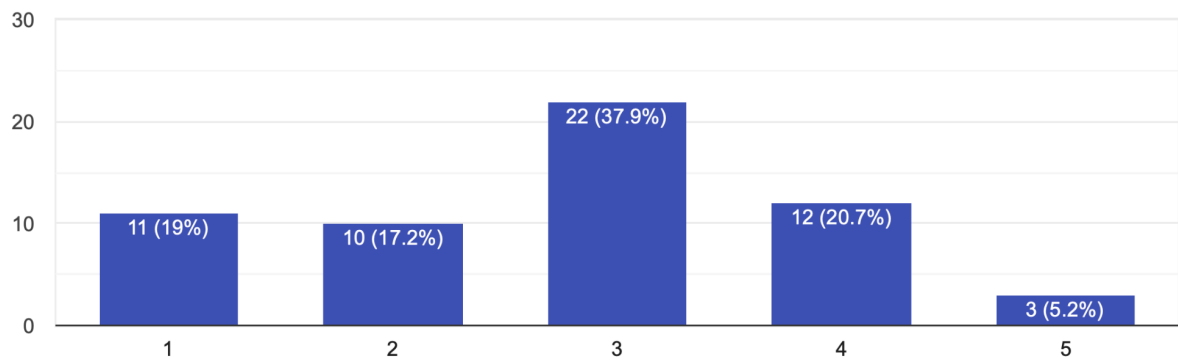


Fig 29: Respondents' view on AI generated content

I believe there is enough transparency in how AI tools make decisions, especially when it comes to fairness

58 responses

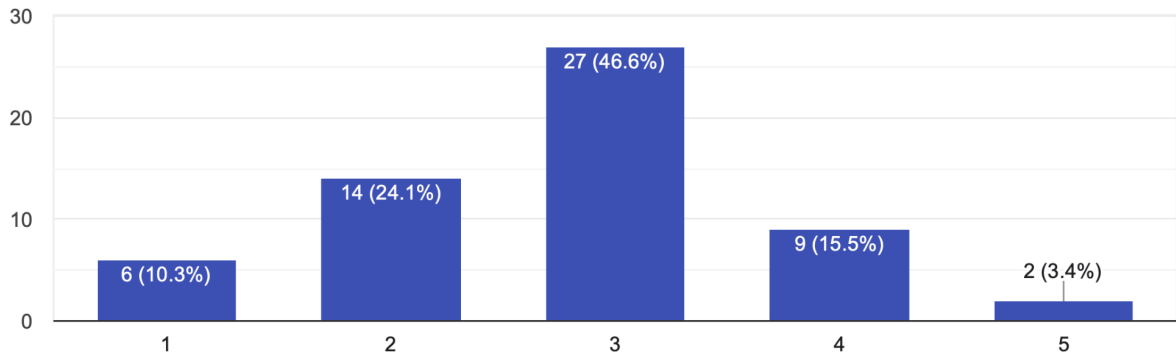


Fig 30: Respondents' view on transparency

Do you think Bangladeshi universities should offer more specialized courses on AI ethics and fairness?

58 responses

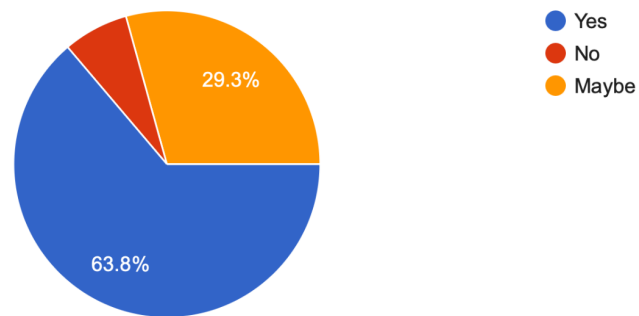


Fig 31: Respondents' view on specialized courses

Trust in Large Language Models (LLM) vs Teachers:

How often do you rely on AI tools for help in understanding complex topics in your coursework?

58 responses

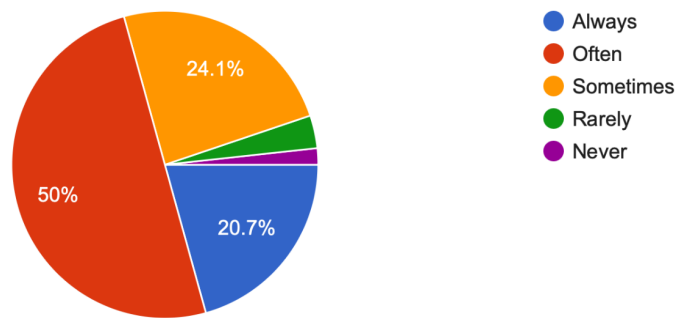


Fig 32: Respondents' reliance on AI tools

When it comes to understanding difficult concepts, which source do you trust more?

58 responses

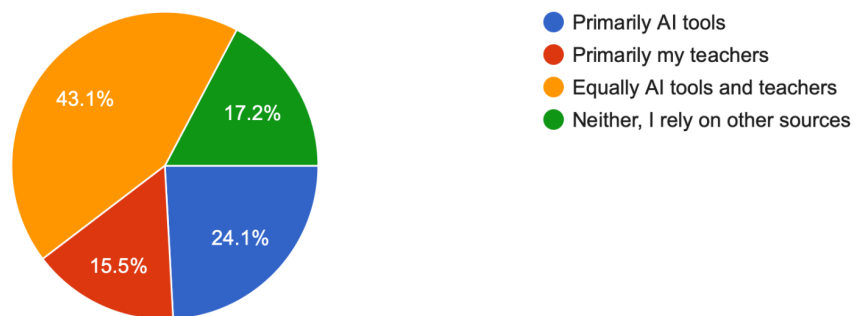


Fig 33: Respondents' source of trust

How confident are you in the accuracy of answers provided by AI tools compared to your teachers' explanations?

58 responses

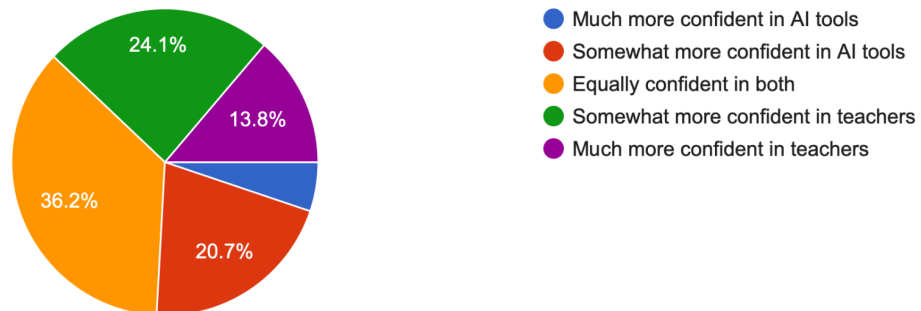


Fig 34: Respondents' confidence in accuracy

If an AI tool and a teacher provided conflicting explanations for the same topic, which one would you be more inclined to believe?

58 responses

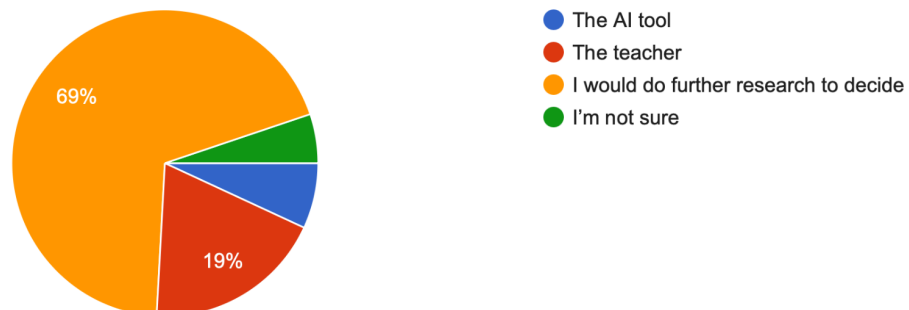


Fig 35: Respondents' inclination to believe

Do you feel that AI tools offer more unbiased information compared to teachers?

58 responses

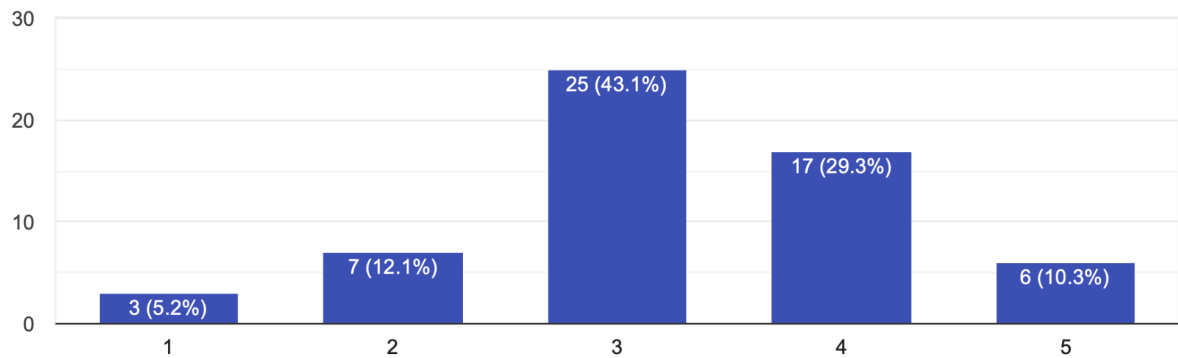


Fig 36: Respondents' view on unbiased information

In your opinion, how well do AI tools understand the context and requirements of your assignments compared to your teachers?

58 responses

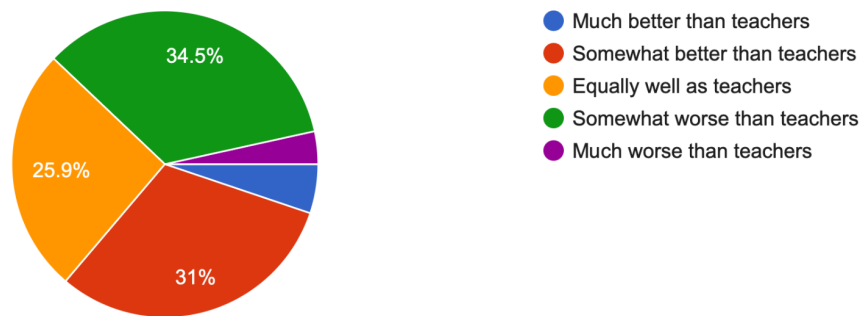


Fig 37: Respondents' view on understanding contexts

Accessibility of AI Tools for Students:

What type of internet access do you use most of the time?

58 responses

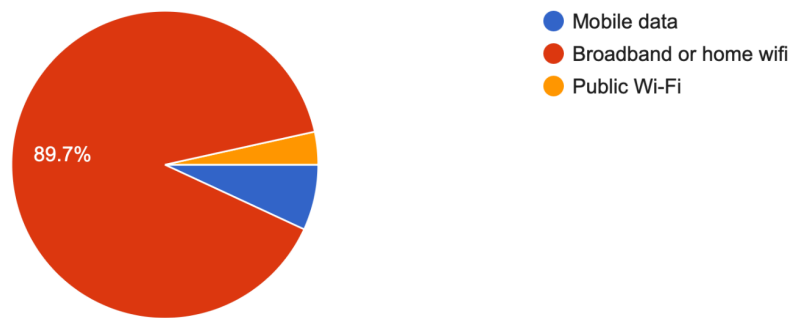


Fig 38: Respondents' internet access type

How often do you face technical issues while using AI tools (e.g., slow network speeds, electricity outages)

58 responses

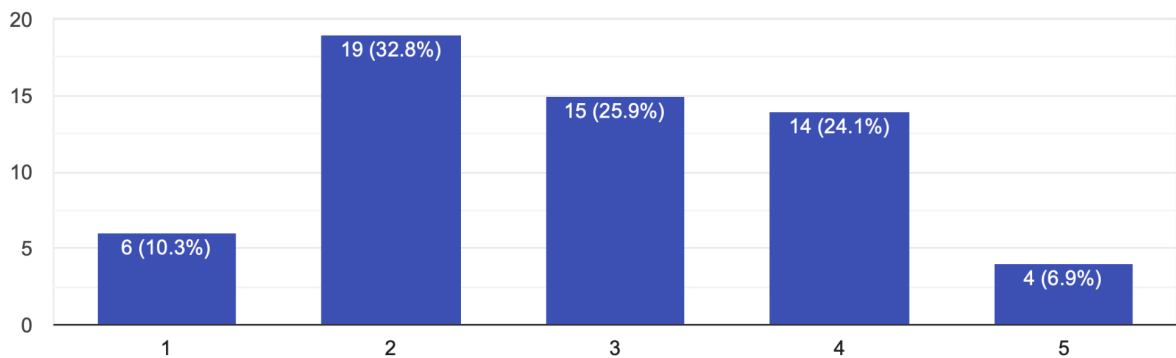


Fig 39: Respondents' facing technical issues

How often does the cost of mobile data or broadband influence your decision to use AI tools?
58 responses

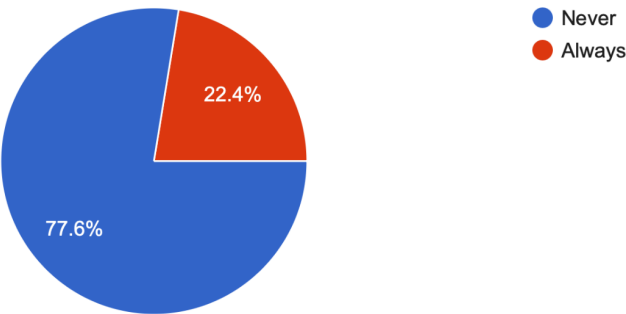


Fig 40: Respondents' view on cost

What type of AI subscription package do you mostly use? (Note : If you are sharing an account with multiple people, that would be considered as a shared account as well)
58 responses

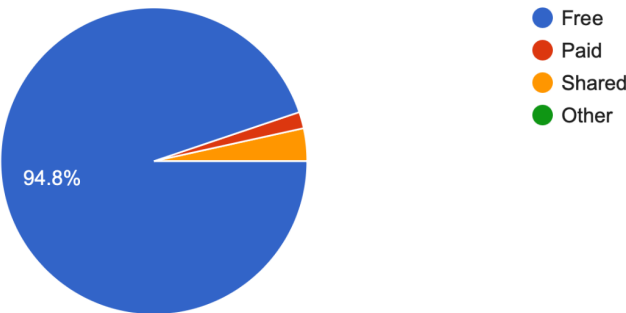


Fig 41: Respondents' AI subscription packages

How affordable do you find premium AI services?

58 responses

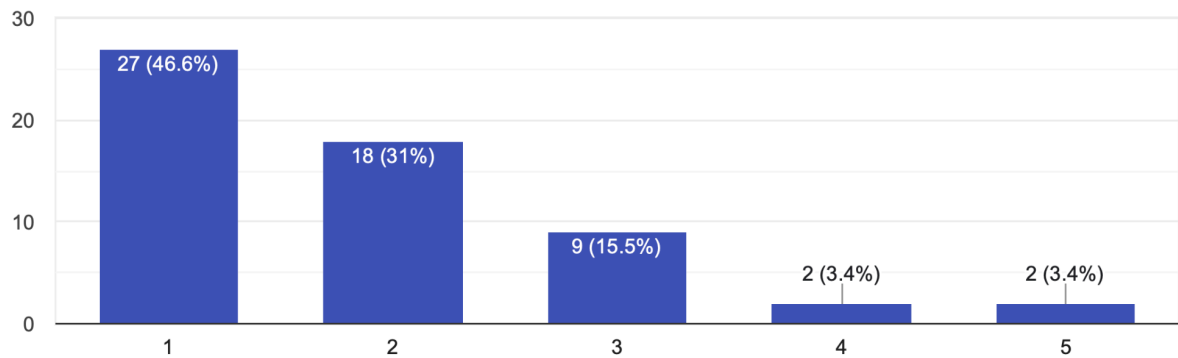


Fig 42: Respondents' view on affordability

How likely are you to use free or open-source AI tools instead of paid alternatives due to affordability issues?

58 responses

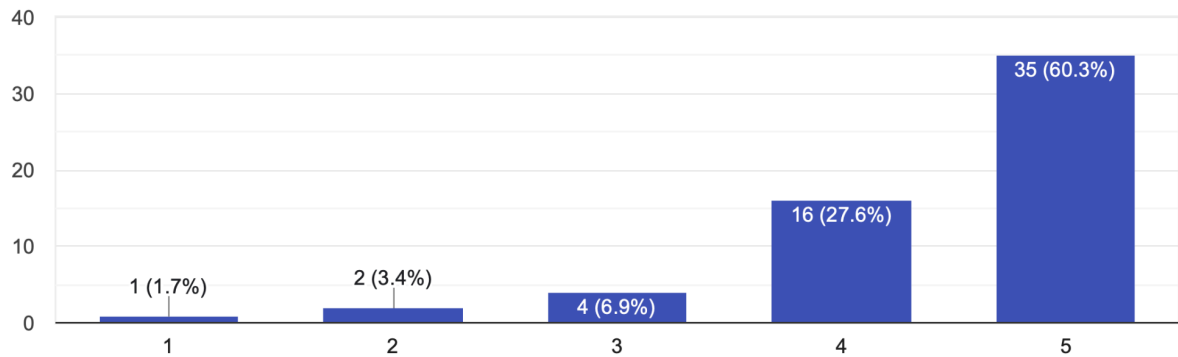


Fig 43: Respondents' likelihood of using free service

How do internet outages affect your ability to work on AI-based assignments or tasks?

58 responses

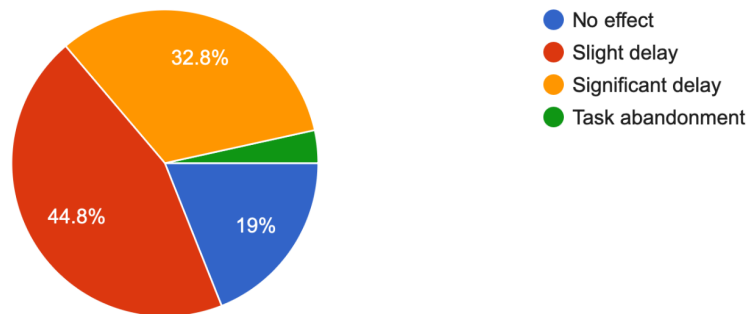


Fig 44: Respondents' response to internet outages

Have you ever avoided using AI tools due to the high cost of internet or data plans?

58 responses

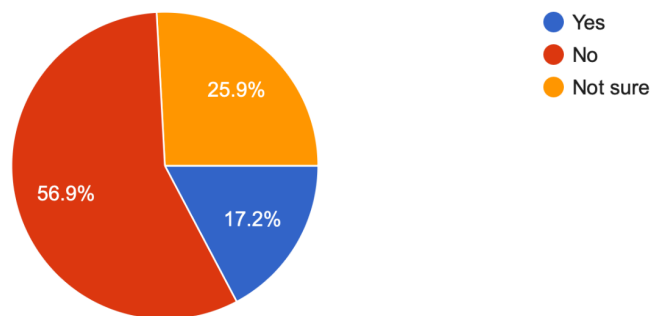


Fig 45: Respondents' avoidance of AI tools

6. Metrics Analysis

Question: I regularly use/used AI for my studies and academic works

D7	fx = CHITEST(D5:D6, E5:E6)				
	A	B	C	D	E
1	I regularly use/used AI for my studies and academic works				
2					
3	Strongly Disagree	3			
4	Disagree	3		Observed	Expected
5	Not sure/ Neutral	9	Agree (ratings 4 and 5)	43	24.5
6	Agree	24	Disagree (ratings 1 and 2)	6	24.5
7	Strongly Agree	19	P value	0.0000001252151994	
8	Total	58			
9					

H_0 : "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement is equal."

This hypothesis focuses on whether students tend to agree or disagree on using AI for academics regularly, ignoring neutrality (rating 3).

We are splitting the observed responses into two groups: agreement and disagreement.

- Observed Frequencies:
 - Agree (ratings 4 and 5): $24+19=43$
 - Disagree (ratings 1 and 2): $3+3=6$
 - Neutral (rating 3): 9 (optional, exclude from this analysis)
- Expected Frequencies: If responses were evenly distributed across agreement and disagreement categories:
 Expected Frequency for Each Group = $(\text{Agree} + \text{Disagree})/2 = 24.5$

Interpretation of the p-value

The p-value $p=0.0000001252151994$ (approximately 1.25×10^{-7}) which is extremely small, much less than the standard significance level ($\alpha=0.05$).

Decision on the Null Hypothesis

Since $p < \alpha$, we **reject the null hypothesis**. This means there is a significant difference between the observed and expected distributions of agreement and disagreement.

Conclusion

Students **do not equally agree and disagree** about regularly using AI for academics. Given the observed data, the majority of students tend to agree (ratings 4 and 5) that they regularly use AI for academic purposes. This indicates a strong inclination toward using AI tools for their studies.

Q2: When searching for something, I tend to use AI tools more than web or book resources

	A	B	C	D	E
1	When searching for something, I tend to use AI tools more than web or book resources				
2					
3	Strongly Disagree	6			
4	Disagree	11		Observed	Expected
5	Not sure/ Neutral	15	Agree (ratings 4 and 5)	26	21.5
6	Agree	17	Disagree (ratings 1 and 2)	17	21.5
7	Strongly Agree	9	P value	0.1699118417	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.17$, which is greater than the significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is not significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is greater than α , we fail to reject the null hypothesis. This means that there is no significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to using AI tools for academics are evenly distributed between agreement and disagreement.

Conclusion:

Based on the results, we can conclude that there is no strong preference or tendency among students for agreeing or disagreeing with the statement about regularly using AI tools for academic purposes. The distribution of responses suggests that students are split in their views, with no significant bias toward either agreeing or disagreeing with the use of AI tools.

Q3: I often use AI for decision-making tasks

	A	B	C	D	E
1	I often use AI for decision-making tasks				
2					
3	Strongly Disagree	14			
4	Disagree	11		Observed	Expected
5	Not sure/ Neutral	16	Agree (ratings 4 and 5)	17	21
6	Agree	14	Disagree (ratings 1 and 2)	25	21
7	Strongly Agree	3	P value	0.2170439078	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.22$ is greater than the significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is not significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is greater than α ($0.22 > 0.05$), we fail to reject the null hypothesis. This means there is no significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the use of AI for decision-making tasks are evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.22, we conclude that there is no significant tendency among students to agree or disagree with the statement that they often use AI for decision-making tasks. The data suggests that students' views on using AI for decision-making are fairly balanced, with no strong preference toward either agreement or disagreement.

Q4: I regularly use AI plugins in the software tools I use (e.g. Copilot in VS Code, Content generating AI in Canva)

	A	B	C	D	E
1	I regularly use AI plugins in the software tools I use (e.g. Copilot in VS Code, Content gen				
2					
3	Strongly Disagree	19			
4	Disagree	14		Observed	Expected
5	Not sure/ Neutral	8	Agree (ratings 4 and 5)	17	25
6	Agree	6	Disagree (ratings 1 and 2)	33	25
7	Strongly Agree	11	P value	0.02365161666	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.023$ is less than the significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.023 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the use of AI plugins in software tools are not evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.023, we conclude that there is a significant tendency among students to either agree or disagree with the statement that they regularly use AI plugins in software tools like Copilot in VS Code or content-generating AI in Canva. The data suggests that students are more likely to either agree or disagree, and their views on using AI plugins in these tools are not evenly balanced.

Motivating Factor for AI:

Q5: The use of AI tools reduced the time for me to understand a topic

	A	B	C	D	E
1	The use of AI tools reduced the time for me to understand a topic				
2					
3	Strongly Disagree	1			
4	Disagree	4		Observed	Expected
5	Not sure/ Neutral	12	Agree (ratings 4 and 5)	41	23
6	Agree	27	Disagree (ratings 1 and 2)	5	23
7	Strongly Agree	14	P value	0.0000001108891117	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.00000011$ is extremely small and much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.00000011 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the use of AI tools in reducing the time it takes to understand a topic are not evenly distributed between agreement and disagreement.

Conclusion:

Given the very small p-value of 0.00000011, we conclude that there is a highly significant tendency among students to agree that the use of AI tools has reduced the time it takes them to understand a topic. The data strongly suggests that AI tools are perceived to have a positive impact on reducing learning time, with a clear preference for agreement over disagreement.

Q6: AI tools enables me to do my work significantly faster

	A	B	C	D	E
1	AI tools enables me to do my work significantly faster				
2					
3	Strongly Disagree	1			
4	Disagree	4		Observed	Expected
5	Not sure/ Neutral	12	Agree (ratings 4 and 5)	41	23
6	Agree	27	Disagree (ratings 1 and 2)	5	23
7	Strongly Agree	14	P value	0.0000001108891117	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.00000011$ is extremely small and much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.00000011 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement that AI tools enable them to do their work significantly faster are not evenly distributed between agreement and disagreement.

Conclusion:

Given the extremely small p-value of 0.00000011, we conclude that there is a highly significant tendency among students to agree that AI tools enable them to complete their work significantly faster. The data strongly suggests that students perceive AI tools as enhancing their work efficiency, with a clear preference for agreement over disagreement.

Q7: There were tasks that I would not have been able to complete without the use of AI tool

	A	B	C	D	E
1	There were tasks that I would not have been able to complete without the use of AI tool				
2					
3	Strongly Disagree	5			
4	Disagree	7		Observed	Expected
5	Not sure/ Neutral	19	Agree (ratings 4 and 5)	27	19.5
6	Agree	20	Disagree (ratings 1 and 2)	12	19.5
7	Strongly Agree	7	P value	0.01630917188	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.016$ is less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.016 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about completing tasks with the help of AI tools are not evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.016, we conclude that there is a significant tendency among students to agree that there were tasks they would not have been able to complete without the use of an AI tool. The data suggests that many students believe AI tools are essential for successfully completing certain tasks, with a clear preference for agreement over disagreement.

The Obstacles of AI Tools

Q8: I have often got wrong/inconsistent answers from AI tools

	A	B	C	D	E
1	I have often got wrong/inconsistent answers from AI tools				
2					
3	Strongly Disagree	1			
4	Disagree	11		Observed	Expected
5	Not sure/ Neutral	22	Agree (ratings 4 and 5)	24	18
6	Agree	16	Disagree (ratings 1 and 2)	12	18
7	Strongly Agree	8	P value	0.0455002639	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.045$ is less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.045 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about receiving wrong or inconsistent answers from AI tools are not evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.045, we conclude that there is a significant tendency among students to either agree or disagree with the statement that they have often received wrong or inconsistent answers from AI tools. The data suggests that there is a noticeable difference in students' experiences with the reliability of AI tools, with a clear preference for either agreeing or disagreeing with the statement, but not being neutral.

Q9: I am concerned that many of the AI tools are storing and tracking my personal information

	A	B	C	D	E
1	I am concerned that many of the AI tools are storing and tracking my personal information				
2					
3	Strongly Disagree	3			
4	Disagree	4		Observed	Expected
5	Not sure/ Neutral	12	Agree (ratings 4 and 5)	39	23
6	Agree	20	Disagree (ratings 1 and 2)	7	23
7	Strongly Agree	19	P value	0.000002380076506	
8	Total	58			
9					
10					

Interpretation of the p-value:

The p-value $p = 0.0000024$ is extremely small and much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.0000024 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the concern about AI tools storing and tracking their personal information are not evenly distributed between agreement and disagreement.

Conclusion:

Given the very small p-value of 0.0000024, we conclude that there is a highly significant

tendency among students to be concerned about AI tools storing and tracking their personal information. The data strongly suggests that students are more likely to express concern about privacy issues with AI tools, with a clear preference for agreement over disagreement.

The Ethical Aspects:

Q10: My institution's academic honesty policy is against AI tools

	A	B	C	D	E
1	My institution's academic honesty policy is against AI tools				
2					
3	Strongly Disagree	5			
4	Disagree	11		Observed	Expected
5	Not sure/ Neutral	30	Agree (ratings 4 and 5)	12	14
6	Agree	8	Disagree (ratings 1 and 2)	16	14
7	Strongly Agree	4	P value	0.449691798	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for institute's policy being pro and against AI is equal"

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for institute's policy being pro and against AI is not equal"

P-value: 0.4497 (greater than 0.05)

So, we do not reject the null hypothesis

Interpretation of the p-value:

The p-value $p = 0.449$ is greater than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is not significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is greater than α ($0.449 > 0.05$), we fail to reject the null hypothesis. This means there is no significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about their institution's academic honesty policy being against AI tools are evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.449, we conclude that there is no significant tendency among students to agree or disagree with the statement that their institution's academic honesty policy is against AI tools. The data suggests that students' views on this matter are fairly balanced, with no strong preference for either agreeing or disagreeing with the policy.

Q11: My teachers/instructors are aware of my use of AI tools in their tasks

	A	B	C	D	E
1	My teachers/instructors are aware of my use of AI tools in their tasks				
2					
3	Strongly Disagree	2			
4	Disagree	7		Observed	Expected
5	Not sure/ Neutral	25	Agree (ratings 4 and 5)	24	16.5
6	Agree	18	Disagree (ratings 1 and 2)	9	16.5
7	Strongly Agree	6	P value	0.009023438818	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for teachers being aware of my use of AI tools in academics is equal"

Alternative Hypothesis (H_1): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for teachers being aware of my use of AI tools in academics is not equal"

Interpretation of the p-value:

The p-value $p = 0.009$ is less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.009 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about their teachers or instructors being aware of their use of AI tools in academic tasks are not evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.009, we conclude that there is a significant tendency among students for either agreeing or disagreeing with the statement that their teachers/instructors are aware of their use of AI tools in academic tasks. The data suggests that students have a strong opinion on this matter, with a clear preference for either agreeing or disagreeing, and it indicates a notable difference in how students perceive their instructors' awareness of their use of AI tools.

Q12: I think that paraphrasing, proof-reading and summarization work for written assignments are fair use cases for AI

	A	B	C	D	E
1	I think that paraphrasing, proof-reading and summarization work for written assignments				
2					
3	Strongly Disagree	2			
4	Disagree	3		Observed	Expected
5	Not sure/ Neutral	19	Agree (ratings 4 and 5)	34	19.5
6	Agree	20	Disagree (ratings 1 and 2)	5	19.5
7	Strongly Agree	14	P value	0.000003421971109	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for perceiving proof-reading and summarization work for written assignments being fair and unfair use cases for AI is equal"

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for perceiving proof-reading and summarization work for written assignments being fair and unfair use cases for AI is not equal"

Interpretation of the p-value:

The p-value $p = 0.0000034$ is extremely small and much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.0000034 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about paraphrasing, proofreading, and summarization being fair use cases for AI are not evenly distributed between agreement and disagreement.

Conclusion:

Given the very small p-value of 0.0000034, we conclude that there is a highly significant tendency among students to agree that paraphrasing, proofreading, and summarization tasks for written assignments are fair use cases for AI. The data strongly suggests that students generally perceive these tasks as appropriate applications for AI tools, with a clear preference for agreement over disagreement.

Q13: I have often used AI tools in some academic activities where it was not allowed

	A	B	C	D	E
1	have often used AI tools in some academic activities where it was not allowed				
2					
3	Strongly Disagree	9			
4	Disagree	15		Observed	Expected
5	Not sure/ Neutral	15	Agree (ratings 4 and 5)	19	21.5
6	Agree	13	Disagree (ratings 1 and 2)	24	21.5
7	Strongly Agree	6	P value	0.4457659079	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for ethical and unethical AI usage in academics is equal"

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement Likelihood for ethical and unethical AI usage in academics is not equal."

Interpretation of the p-value:

The p-value $p = 0.446$ is greater than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is not significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is greater than α ($0.446 > 0.05$), we fail to reject the null hypothesis. This means there is no significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about using AI tools in academic activities where it was not allowed are evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.446, we conclude that there is no significant tendency among students to agree or disagree with the statement that they have often used AI tools in academic activities where it was not allowed. The data suggests that students are evenly split on this matter, with no clear preference for either agreeing or disagreeing with the use of AI tools in situations where they were not allowed.

Q14: I was often penalized for the use of AI tools in academic activities

	A	B	C	D	E
1	I was often penalized for the use of AI tools in academic activities				
2					
3	Strongly Disagree	27			
4	Disagree	10		Observed	Expected
5	Not sure/ Neutral	16	Agree (ratings 4 and 5)	5	21
6	Agree	3	Disagree (ratings 1 and 2)	37	21
7	Strongly Agree	2	P value	0.0000007904640392	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for being penalized and not being penalized for using AI tools in academics is equal."

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement Likelihood for being penalized and not being penalized for using AI tools in academics is not equal."

Interpretation of the p-value:

The p-value $p = 0.00000079$ is extremely small and much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.00000079 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about being penalized for using AI tools in academic activities are not evenly distributed between agreement and disagreement.

Conclusion:

Given the very small p-value of 0.00000079 , we conclude that there is a highly significant tendency among students to disagree that they were often penalized for using AI tools in academic activities. The data strongly suggests that students do not perceive penalties for AI tool use as a notable issue, with a clear preference for disagreement over agreement.

Trust in Large Language Models (LLM) vs Teachers

Q15: Do you feel that AI tools offer more unbiased information compared to teachers?

	A	B	C	D	E
1	Do you feel that AI tools offer more unbiased information compared to teachers?				
2					
3	Strongly Disagree	3			
4	Disagree	7		Observed	Expected
5	Not sure/ Neutral	25	Agree (ratings 4 and 5)	23	16.5
6	Agree	17	Disagree (ratings 1 and 2)	10	16.5
7	Strongly Agree	6	P value	0.02363510198	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement 'AI tools offer more unbiased information compared to teachers' is equal."

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement 'AI tools offer more unbiased information compared to teachers' is not equal."

Interpretation of the p-value:

The p-value $p = 0.023$ is less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.023 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about AI tools offering more unbiased information compared to teachers are not evenly distributed between agreement and disagreement.

Conclusion:

Given the p-value of 0.023, we conclude that there is a significant tendency among students to either agree or disagree with the statement that AI tools offer more unbiased information compared to teachers. The data suggests that a notable proportion of students have strong opinions on the relative bias of AI tools versus teachers, with a clear preference for either agreeing or disagreeing, but not remaining neutral.

Accessibility of AI Tools for Students

Q16: How affordable do you find premium AI services?

	A	B	C	D	E
1	How affordable do you find premium AI services?				
2					
3	Strongly Disagree	27			
4	Disagree	18		Observed	Expected
5	Not sure/ Neutral	9	Agree (ratings 4 and 5)	4	24.5
6	Agree	2	Disagree (ratings 1 and 2)	45	24.5
7	Strongly Agree	2	P value	0.0000000047089804	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement 'Premium AI tools are affordable '"

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement 'Premium AI tools are not affordable ."

Interpretation of the p-value:

The p-value $p = 0.0000000047$ is extremely small and much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is less than α ($0.0000000047 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about the affordability of premium AI services are not evenly distributed between agreement and disagreement.

Conclusion:

Given the extremely small p-value of 0.0000000047 , we conclude that there is a highly significant tendency among students to disagree with the statement that premium AI

services are affordable. The data strongly suggests that most students perceive premium AI services as unaffordable, with a clear preference for disagreement over agreement.

Q17: How likely are you to use free or open-source AI tools instead of paid alternatives due to affordability issues

	A	B	C	D	E
1	How likely are you to use free or open-source AI tools instead of paid alternatives due to affordability issues?				
2					
3	Strongly Disagree	1			
4	Disagree	2		Observed	Expected
5	Not sure/ Neutral	4	Agree (ratings 4 and 5)	51	27
6	Agree	16	Disagree (ratings 1 and 2)	3	27
7	Strongly Agree	35	P value	0	
8	Total	58			
9					
10					

H₀ (Null Hypothesis): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for open source AI tools and paid alternative usage is equal."

Alternative Hypothesis (H₁): "The proportion of students agreeing (Likert scale ratings 4 and 5) and disagreeing (Likert scale ratings 1 and 2) with the statement "Likelihood for open source AI tools and paid alternative usage is not equal."

Interpretation of the p-value:

The p-value is so small that Excel rounds it to 0, which means it is much less than the standard significance level ($\alpha = 0.05$). This indicates that the observed distribution of agreement and disagreement is significantly different from the expected distribution.

Decision on the Null Hypothesis:

Since the p-value is effectively 0, which is less than α ($0 < 0.05$), we reject the null hypothesis. This means there is a significant difference between the observed and expected proportions of agreement and disagreement. Therefore, we conclude that students' responses to the statement about using free or open-source AI tools due to affordability are not evenly distributed between agreement and disagreement.

Conclusion:

Given the extremely small p-value (close to 0), we conclude that there is a highly

significant tendency among students to **agree or strongly agree** with the statement that they are more likely to use free or open-source AI tools due to affordability issues. The data strongly suggests that students are highly motivated by affordability when choosing between free/open-source and paid AI tools, with a clear preference for agreeing or strongly agreeing with the use of free alternatives.

7. Final Thoughts and Implications

The survey results reflect the growing role of AI tools in students' academic lives, underscoring both their widespread adoption and the challenges they bring. Students are increasingly turning to AI tools to enhance their academic work, particularly for tasks like decision-making, summarization, and time-saving. However, concerns about the affordability of premium AI services stand out, with many students expressing a preference for free or open-source alternatives due to financial constraints. At the same time, students hold mixed views on the reliability and bias of AI-generated content, with some questioning whether AI tools can offer unbiased information compared to human teachers. The survey also revealed that AI tools are being used in academic tasks where they may not be explicitly permitted, and some students have faced penalties as a result. Despite this, there is a strong inclination to view certain uses of AI, such as paraphrasing and proofreading, as acceptable. Additionally, a noticeable gap in awareness between students and teachers regarding the use of AI tools suggests a need for better communication and more explicit policies surrounding AI usage. Overall, while AI tools are seen as valuable educational aids, there is a clear need for institutions to address concerns around accessibility, ethical use, and academic integrity to ensure these tools are used responsibly and effectively.